Results

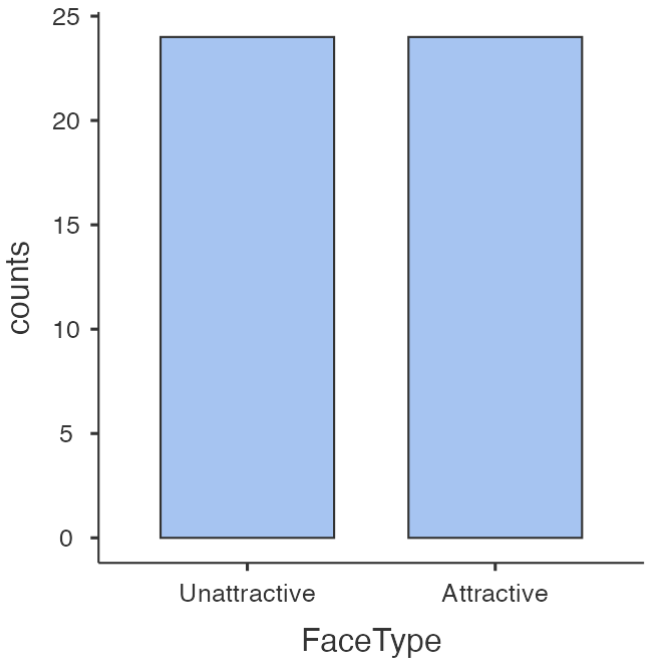
Descriptives

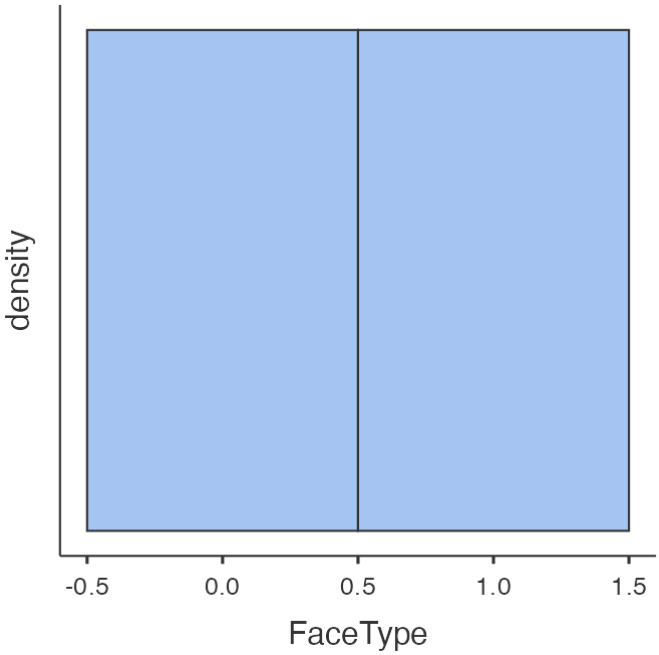
Descriptives

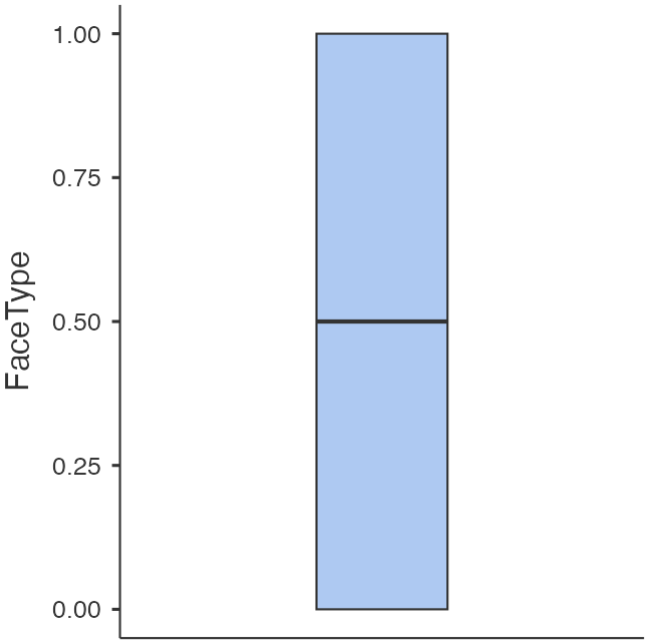
	FaceType	Alcohol	Attractiveness
N	48	48	48
Missing	0	0	0
Mean	0.500	1.00	5.67
Median	0.500	1.00	6.00
Standard deviation	0.505	0.825	1.59
Minimum	0	0	1.00
Maximum	1	2	8.00
Skewness	0.00	0.00	-0.783
Std. error skewness	0.343	0.343	0.343
Kurtosis	-2.09	-1.53	0.665
Std. error kurtosis	0.674	0.674	0.674
Shapiro-Wilk W	0.637	0.794	0.926
Shapiro-Wilk p	<.001	<.001	0.005

Plots

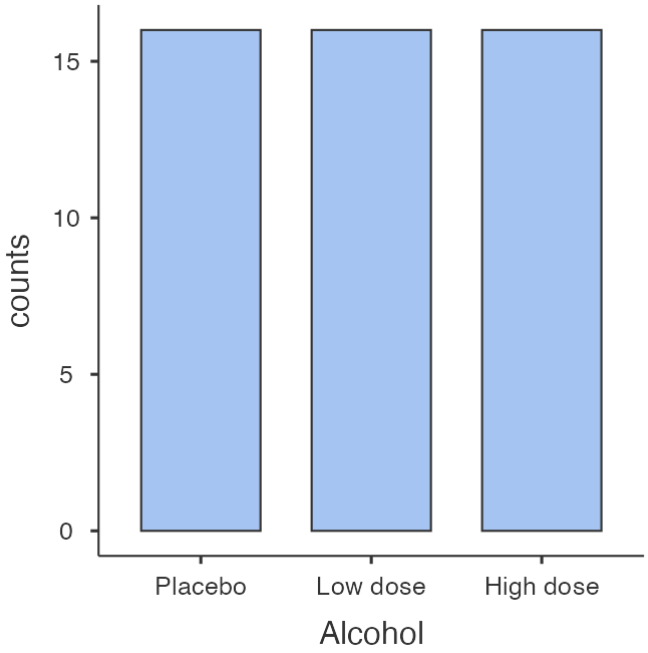
FaceType

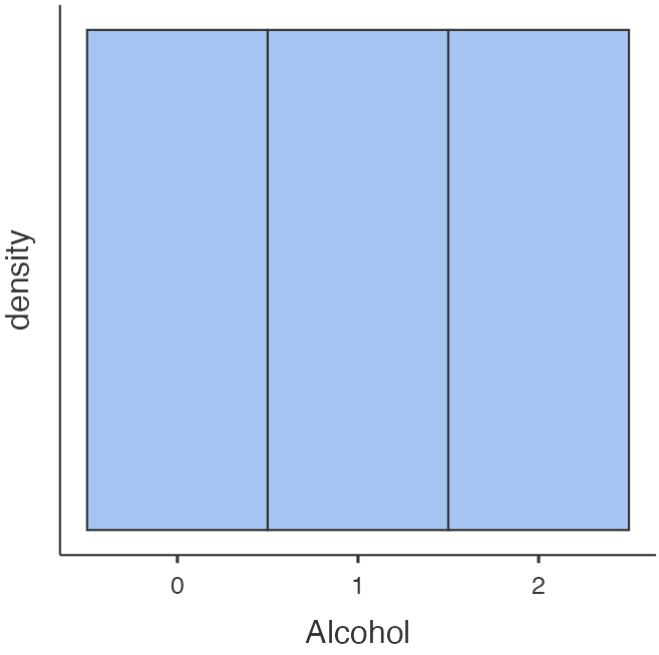


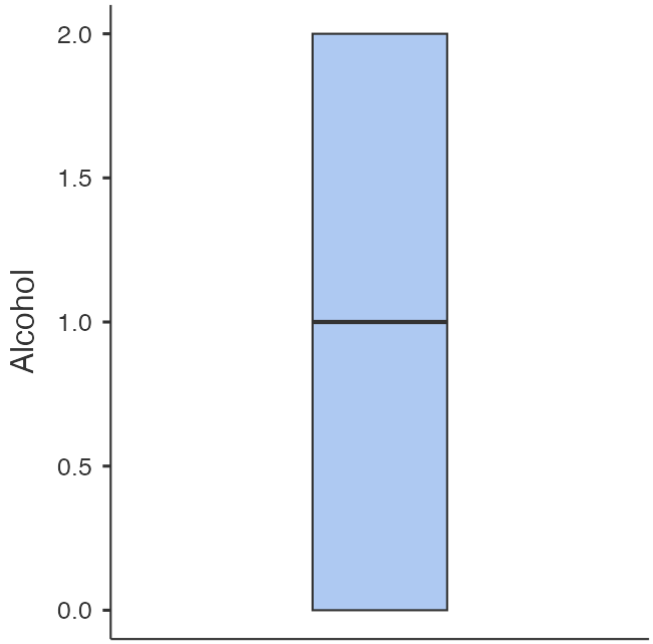




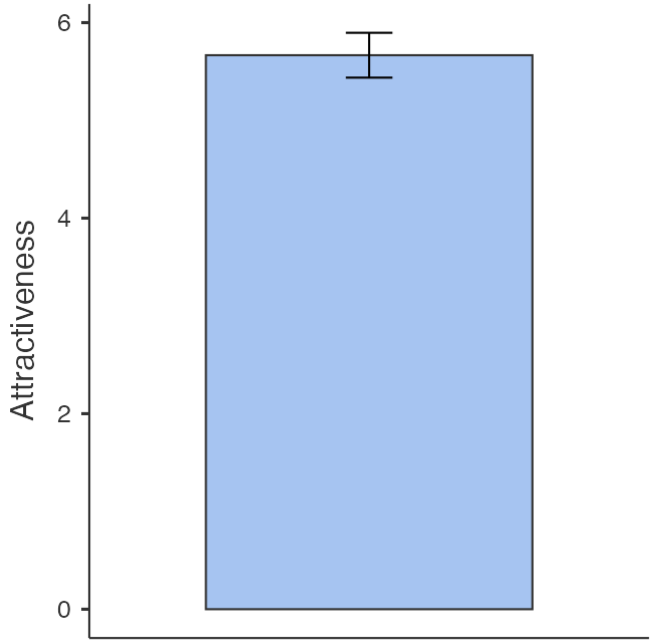
Alcohol

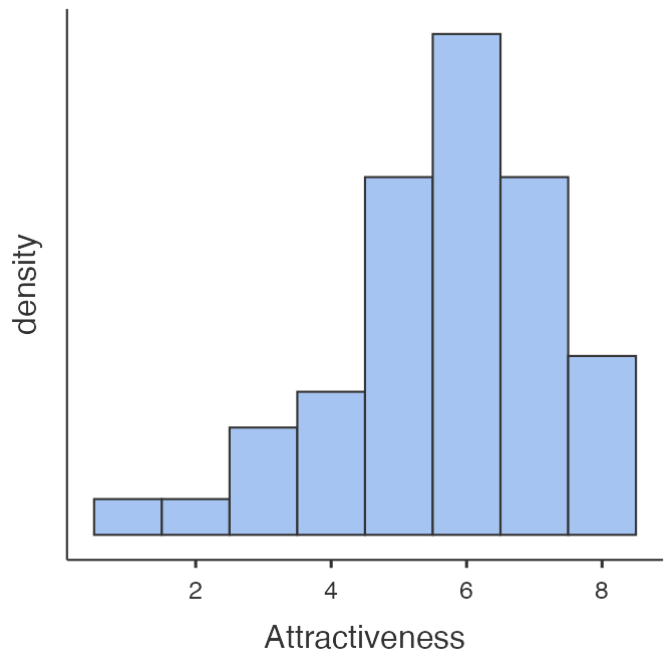

counts
15
10
5
0
Placebo
Low dose
High dose
Alcohol

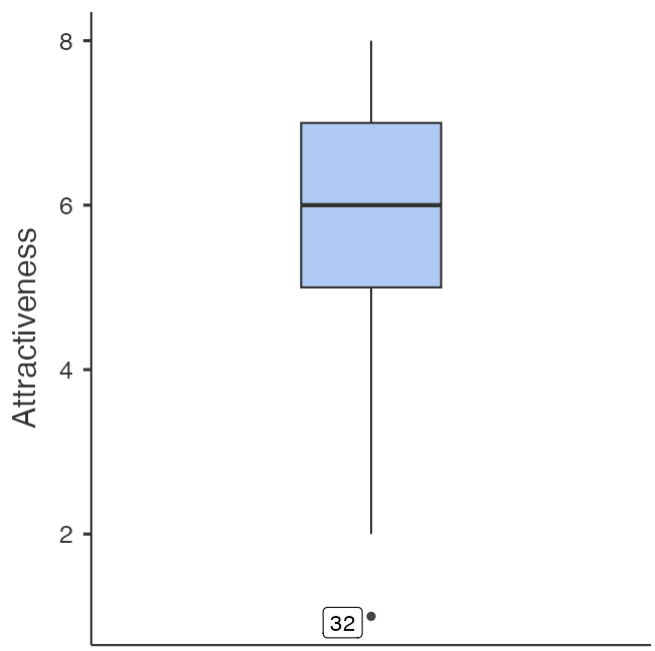

density
0
1
2
Alcohol



Attractiveness







Relationships, Prediction, and Group Comparisons

Welcome to Statkat! This tool will help you to find an appropriate statistical method given the measurement level of your data. Make sure you have correctly defined the measurement levels of your variables on the Data tab. You can change the measurement level of a variable via the Setup button on the Data tab, or by double clicking on a column header of interest. You have selected the Relationships, Prediction, and Group Comparisons option. This is the place to be if you are interested in

- the relationship between two or more variables, or
- predicting one variable from other variables, or
- the difference between independent (unrelated) groups on a certain variable.

To get started, drop a variable in the box below Variable 1 / Dependent Variable, and one or more variables in the box below Variable 2 / Independent Variables. Our tool will then come up with a statistical method that may be appropriate for your data! In addition, you can drop one or more variables in the box below Control Variables. Control variables are variables that you are not particularly interested in, but which may be related to the dependent variable and possibly also to the independent variables. In experiments (with random assignment), control variables are often included to increase power. In observational studies, control variables are often included mainly to equate subjects on the control variables. This prevents the control variables from

confounding the relationships between the independent variables and the dependent variable. If your research question does not make a clear distinction between an independent variable and a dependent variable, the decision of which variable to define as Variable 1/Dependent Variable and which as Variable 2/Independent Variables can be arbitrary. But doesn't this decision affect the recommended method? Well, in some cases it does affect the primary method recommendation, but if a simpler method can be performed by flipping the two variables, this is usually mentioned. It is then up to you which of the recommended methods you prefer. It is important to keep in mind here that none of the correlational statistical techniques can say anything about causality anyway (not even a method like regression analysis), so even if you do make a distinction between an independent and dependent variable, the statistical method will only say something about association, not causation. Note: Our advice is based on the measurement level of your data and on the number of variables entered. There can be details related to your data, task, or assignment that may render the advice moot. Always check the assumptions made by the statistical method before interpreting the results. We always try to come up with the least complicated method that might be applicable given your data. Keep in mind that there may be other, more advanced, methods that might be applicable as well.

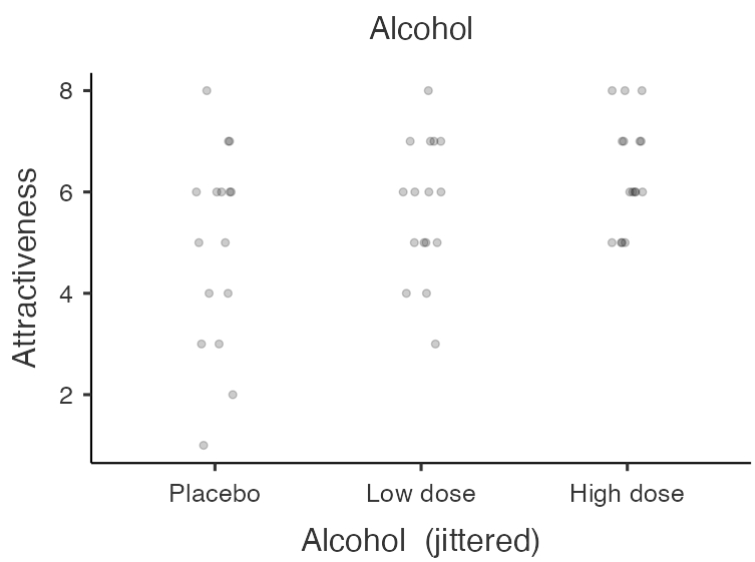
Relationships, Prediction, and Group Comparisons

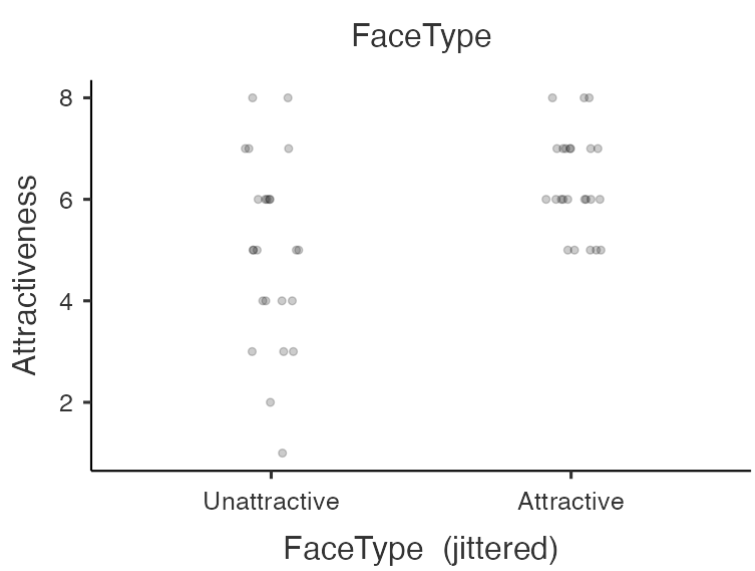
You have entered a numeric dependent variable and two categorical (nominal/ordinal) independent variables. Hence, a two way ANOVA seems to be a good option for you! In order to run this analysis in jamovi, go to: ANOVA > ANOVA

- Drop your numeric dependent variable in the box below Dependent Variable and your two independent (grouping) variables in the box below Fixed Factors

Click on the link to learn more about this method!

Scatter Plots of Bivariate Relationships - Dependent/Independent Variables





ANOVA

ANOVA - Attractiveness

	Sum of Squares	df	Mean Square	F	p	ω²
Alcohol	16.5	2	8.27	6.04	0.005	0.115
FaceType	21.3	1	21.33	15.58	<.001	0.166
Alcohol ✻ FaceType	23.3	2	11.65	8.51	<.001	0.171
Residuals	57.5	42	1.37			

[3]

Assumption Checks

Homogeneity of Variances Tests

	Statistic	df	df2	p
Levene's	0.702	5	42	0.625
Bartlett's	3.14	5		0.678

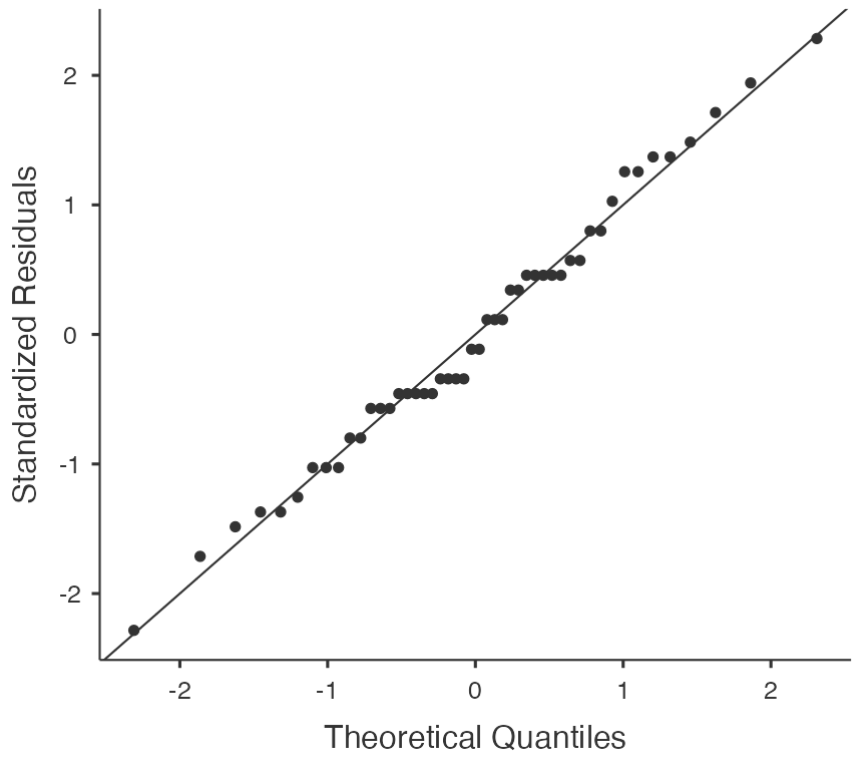
Note. Additional results provided by *moretests*

Normality tests

	statistic	p
Shapiro-Wilk	0.987	0.878
Kolmogorov-Smirnov	0.112	0.585
Anderson-Darling	0.288	0.605

Note. Additional results provided by *moretests*

Q-Q Plot



ANCOVA

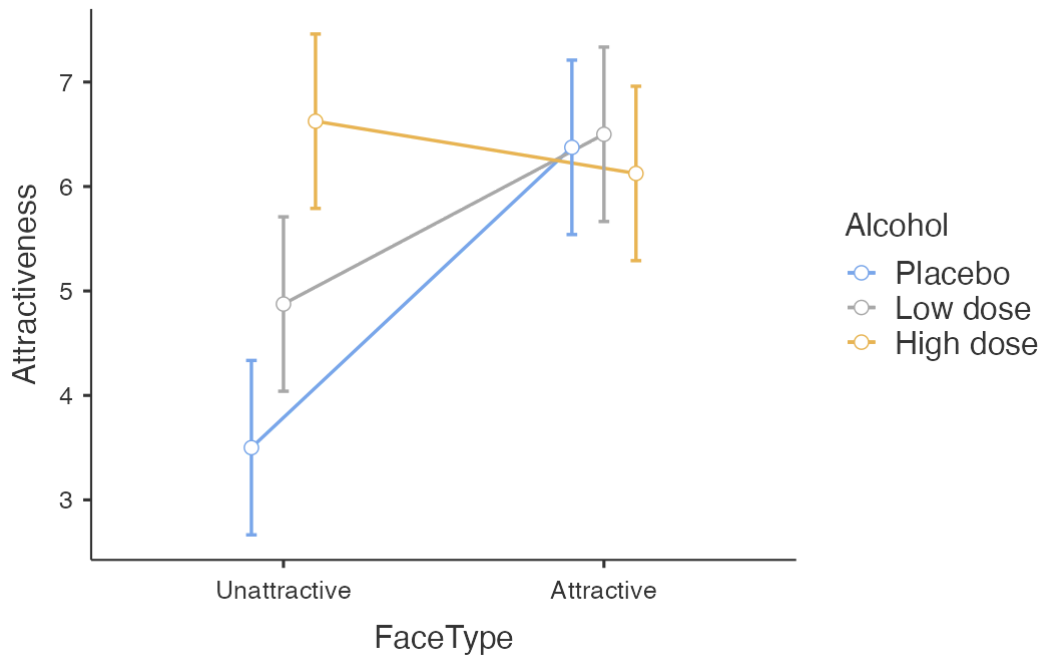
ANCOVA - Attractiveness

	Sum of Squares	df	Mean Square	F	p
Alcohol	16.5	2	8.27	6.04	0.005
FaceType	21.3	1	21.33	15.58	<.001
Alcohol ✻ FaceType	23.3	2	11.65	8.51	<.001
Residuals	57.5	42	1.37		

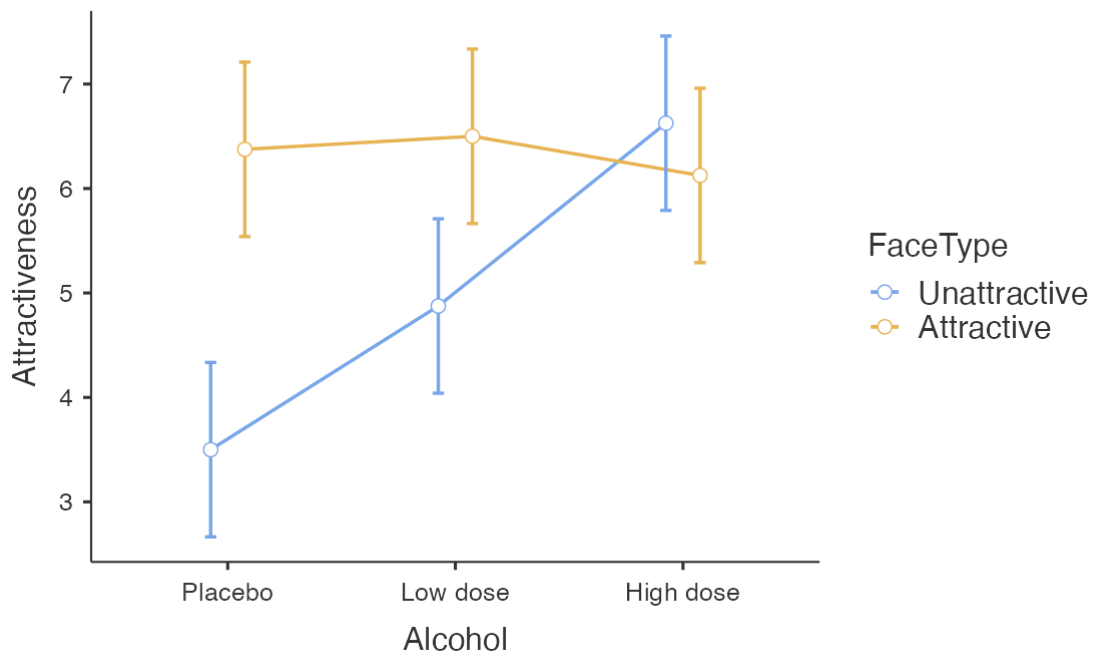
[3]

Estimated Marginal Means

FaceType ✻ Alcohol



Alcohol ✻ FaceType



[4]

References

[1] The jamovi project (2024). *jamovi*. (Version 2.6) [Computer Software]. Retrieved from https://www.jamovi.org.

[2] R Core Team (2024). *R: A Language and environment for statistical computing*. (Version 4.4) [Computer software]. Retrieved from https://cran.r-project.org. (R packages retrieved from CRAN snapshot 2024-08-07).

[3] Fox, J., & Weisberg, S. (2023). *car: Companion to Applied Regression*. [R package]. Retrieved from https://cran.r-project.org/package=car.

[4] Lenth, R. (2023). *emmeans: Estimated Marginal Means, aka Least-Squares Means*. [R package]. Retrieved from https://cran.r-project.org/package=emmeans.